

Weak Optimal Transport with a Barycentric Projection Cost

1 Continuous weak optimal transport

Let

$$\alpha \in \mathcal{P}_2(\mathbb{R}^d), \quad \beta \in \mathcal{P}_2(\mathbb{R}^d)$$

be probability measures with finite second moments. The set of couplings between α and β is

$$\Pi(\alpha, \beta) = \left\{ \pi \in \mathcal{P}(\mathbb{R}^d \times \mathbb{R}^d) : (\text{proj}_x)_\# \pi = \alpha, (\text{proj}_y)_\# \pi = \beta \right\}.$$

Since the first marginal of π is α , one may disintegrate π as

$$\pi(dx, dy) = \alpha(dx) \pi_x(dy),$$

where π_x is the conditional distribution of Y given $X = x$. Its conditional barycenter is

$$b_\pi(x) := \int_{\mathbb{R}^d} y \pi_x(dy) = \mathbb{E}_\pi[Y \mid X = x].$$

The quadratic weak optimal transport cost is

$$\mathcal{W}_{\text{weak}}^2(\alpha, \beta) := \inf_{\pi \in \Pi(\alpha, \beta)} \int_{\mathbb{R}^d} \|x - b_\pi(x)\|^2 \alpha(dx).$$

Equivalently, introducing the weak cost

$$C(x, p) = \left\| x - \int_{\mathbb{R}^d} y p(dy) \right\|^2,$$

one has

$$\mathcal{W}_{\text{weak}}^2(\alpha, \beta) = \inf_{\pi \in \Pi(\alpha, \beta)} \int_{\mathbb{R}^d} C(x, \pi_x) \alpha(dx).$$

This differs from classical quadratic optimal transport:

$$W_2^2(\alpha, \beta) = \inf_{\pi \in \Pi(\alpha, \beta)} \int_{\mathbb{R}^d \times \mathbb{R}^d} \|x - y\|^2 \pi(dx, dy).$$

Indeed, conditional Jensen's inequality gives

$$\|x - \mathbb{E}[Y \mid X = x]\|^2 \leq \mathbb{E}[\|x - Y\|^2 \mid X = x],$$

and consequently

$$\mathcal{W}_{\text{weak}}^2(\alpha, \beta) \leq W_2^2(\alpha, \beta).$$

Moreover,

$$\mathcal{W}_{\text{weak}}^2(\alpha, \beta) = 0$$

whenever there exists a martingale coupling satisfying

$$\mathbb{E}[Y \mid X] = X.$$

In particular, by Strassen's theorem, this happens when α is dominated by β in convex order.

2 Discrete formulation

Assume that

$$\alpha = \sum_{i=1}^n a_i \delta_{x_i}, \quad \beta = \sum_{j=1}^m b_j \delta_{y_j},$$

where

$$a_i > 0, \quad b_j > 0, \quad \sum_{i=1}^n a_i = \sum_{j=1}^m b_j = 1.$$

A discrete coupling is represented by a nonnegative transport matrix

$$P = (P_{ij}) \in \mathbb{R}_+^{n \times m}.$$

The marginal constraints are

$$P \mathbf{1}_m = \mathbf{a}, \quad P^\top \mathbf{1}_n = \mathbf{b},$$

so the discrete transport polytope is

$$U(\mathbf{a}, \mathbf{b}) := \left\{ P \in \mathbb{R}_+^{n \times m} : P \mathbf{1}_m = \mathbf{a}, P^\top \mathbf{1}_n = \mathbf{b} \right\}.$$

The conditional probability of y_j given x_i is

$$\mathbb{P}(Y = y_j \mid X = x_i) = \frac{P_{ij}}{a_i}.$$

Thus, the barycentric projection of row i is

$$\bar{y}_i(P) := \mathbb{E}[Y \mid X = x_i] = \frac{1}{a_i} \sum_{j=1}^m P_{ij} y_j.$$

The discrete weak transport problem is therefore

$$\min_{P \in U(\mathbf{a}, \mathbf{b})} \mathcal{B}(P), \quad \mathcal{B}(P) := \sum_{i=1}^n a_i \left\| x_i - \frac{1}{a_i} \sum_{j=1}^m P_{ij} y_j \right\|^2.$$

An equivalent expression is

$$\mathcal{B}(P) = \sum_{i=1}^n \frac{1}{a_i} \left\| a_i x_i - \sum_{j=1}^m P_{ij} y_j \right\|^2.$$

Notice that the objective depends on each row of P only through its barycenter. Consequently, two different transport matrices can have exactly the same weak transport cost.

3 Convexity

For fixed positive weights a_i , the map

$$P \mapsto \frac{1}{a_i} \left\| a_i x_i - \sum_j P_{ij} y_j \right\|^2$$

is a convex quadratic function of the i -th row of P . Hence $\mathcal{B}$ is convex.

More explicitly, for a perturbation $H \in \mathbb{R}^{n \times m}$,

$$D^2\mathcal{B}(P)[H, H] = 2 \sum_{i=1}^n \frac{1}{a_i} \left\| \sum_{j=1}^m H_{ij} y_j \right\|^2 \geq 0.$$

Since $U(\mathbf{a}, \mathbf{b})$ is convex and defined by linear constraints, the discrete weak transport problem is a convex optimization problem.

It is generally not strictly convex. Indeed, if

$$\sum_{j=1}^m H_{ij} y_j = 0 \quad \text{for every } i,$$

then the quadratic form above vanishes, even when $H \neq 0$. Thus, different couplings with identical row barycenters may all be minimizers.

The same convexity principle holds in the continuous setting because

$$p \mapsto \left\| x - \int y p(dy) \right\|^2$$

is convex in the conditional measure p .

4 KL-regularized weak transport

A natural entropic regularization uses the independent coupling $\alpha \otimes \beta$ as reference:

$$\inf_{\pi \in \Pi(\alpha, \beta)} \left\{ \int \|x - b_\pi(x)\|^2 \alpha(dx) + \varepsilon \text{KL}(\pi \| \alpha \otimes \beta) \right\}.$$

In the discrete setting, define

$$Q_{ij} := a_i b_j$$

and use the generalized KL divergence

$$\text{KL}(P \| Q) = \sum_{i=1}^n \sum_{j=1}^m \left[P_{ij} \log \left(\frac{P_{ij}}{Q_{ij}} \right) - P_{ij} + Q_{ij} \right].$$

The regularized problem is

$$\min_{P \in U(\mathbf{a}, \mathbf{b})} \left\{ \mathcal{B}(P) + \varepsilon \text{KL}(P \| \mathbf{a} \mathbf{b}^\top) \right\}.$$

For $\varepsilon > 0$, the KL term is strictly convex on positive matrices. Thus, under the assumption $a_i, b_j > 0$, the regularized problem has a unique optimal transport matrix.

5 Optimality conditions

Let

$$\bar{y}_i(P) = \frac{1}{a_i} \sum_j P_{ij} y_j.$$

The gradient of the barycentric cost is

$$\frac{\partial \mathcal{B}}{\partial P_{ij}}(P) = 2 \langle \bar{y}_i(P) - x_i, y_j \rangle.$$

Define the endogenous cost matrix

$$C_{ij}(P) := 2 \langle \bar{y}_i(P) - x_i, y_j \rangle.$$

The first-order optimality conditions imply that there exist potentials f_i and g_j such that

$$P_{ij} = a_i b_j \exp \left(\frac{f_i + g_j - C_{ij}(P)}{\varepsilon} \right).$$

Unlike classical entropic optimal transport, the cost matrix depends on the unknown coupling through its row barycenters. Therefore, one does not obtain the solution from a single Sinkhorn scaling. Instead, the solution satisfies a nonlinear Sinkhorn fixed-point equation.

6 Saddle-point reformulation

Use the Fenchel identity

$$\|x - z\|^2 = \sup_{u \in \mathbb{R}^d} \{2\langle u, x - z \rangle - \|u\|^2\}.$$

Applying it row by row yields

$$\mathcal{B}(P) = \sup_{u_1, \dots, u_n \in \mathbb{R}^d} \left\{ \sum_{i=1}^n (2a_i \langle u_i, x_i \rangle - a_i \|u_i\|^2) - 2 \sum_{i,j} P_{ij} \langle u_i, y_j \rangle \right\}.$$

Consequently, the regularized weak transport problem is equivalent to

$$\sup_{\mathbf{u}} \left\{ \sum_i (2a_i \langle u_i, x_i \rangle - a_i \|u_i\|^2) + \min_{P \in U(\mathbf{a}, \mathbf{b})} \left[\sum_{i,j} C_{ij}(\mathbf{u}) P_{ij} + \varepsilon \text{KL}(P \| \mathbf{a} \mathbf{b}^\top) \right] \right\},$$

where

$$C_{ij}(\mathbf{u}) = -2\langle u_i, y_j \rangle.$$

For fixed $\mathbf{u}$, the inner problem is now an ordinary entropic linear optimal transport problem and can be solved by Sinkhorn iterations.

The corresponding Gibbs kernel is

$$K_{ij}(\mathbf{u}) = \exp \left(-\frac{C_{ij}(\mathbf{u})}{\varepsilon} \right) = \exp \left(\frac{2\langle u_i, y_j \rangle}{\varepsilon} \right).$$

The optimal coupling for fixed $\mathbf{u}$ has the scaling form

$$P_{ij} = a_i b_j r_i K_{ij}(\mathbf{u}) s_j,$$

where $r_i, s_j > 0$ enforce the marginal constraints.

The Sinkhorn scaling iterations are

$$r_i \leftarrow \frac{1}{\sum_{j=1}^m b_j K_{ij}(\mathbf{u}) s_j}, \quad i = 1, \dots, n,$$

followed by

$$s_j \leftarrow \frac{1}{\sum_{i=1}^n a_i r_i K_{ij}(\mathbf{u})}, \quad j = 1, \dots, m.$$

Let $P(\mathbf{u})$ denote the result of this inner Sinkhorn problem, and let

$$\bar{y}_i(\mathbf{u}) := \frac{1}{a_i} \sum_j P_{ij}(\mathbf{u}) y_j.$$

By the envelope theorem, the gradient of the outer concave objective $\Phi(\mathbf{u})$ is

$$\nabla_{u_i} \Phi(\mathbf{u}) = 2a_i (x_i - u_i - \bar{y}_i(\mathbf{u})).$$

At the optimum,

$$\boxed{u_i^* = x_i - \bar{y}_i(P^*)}.$$

Substituting this identity into the Gibbs kernel recovers the nonlinear optimality condition

$$K_{ij}^* = \exp \left(\frac{2\langle x_i - \bar{y}_i(P^*), y_j \rangle}{\varepsilon} \right).$$

7 Sinkhorn-type algorithm

A practical algorithm alternates between a Sinkhorn solve for fixed $\mathbf{u}$ and a damped ascent step in $\mathbf{u}$.

Algorithm 1 Damped Sinkhorn iteration for weak entropic transport

- 1: Initialize $u_i^{(0)} = 0$ and $s_j^{(0)} = 1$.
- 2: **for** $k = 0, 1, 2, \dots$ **do**
- 3: Form the kernel

$$K_{ij}^{(k)} = \exp \left(\frac{2\langle u_i^{(k)}, y_j \rangle}{\varepsilon} \right).$$

- 4: Using warm starts, iterate

$$r_i \leftarrow \frac{1}{\sum_j b_j K_{ij}^{(k)} s_j}, \quad s_j \leftarrow \frac{1}{\sum_i a_i r_i K_{ij}^{(k)}}$$

until the marginal errors are sufficiently small.

- 5: Set

$$P_{ij}^{(k)} = a_i b_j r_i K_{ij}^{(k)} s_j.$$

- 6: Compute the barycentric projections

$$\bar{y}_i^{(k)} = \frac{1}{a_i} \sum_j P_{ij}^{(k)} y_j.$$

- 7: Update, with a damping parameter $\rho_k > 0$,

$$u_i^{(k+1)} = u_i^{(k)} + \rho_k \left(x_i - u_i^{(k)} - \bar{y}_i^{(k)} \right).$$

- 8: **end for**
-

The update is a diagonally preconditioned gradient-ascent step for the concave outer objective. The undamped fixed-point choice $\rho_k = 1$ gives

$$u_i^{(k+1)} = x_i - \bar{y}_i^{(k)},$$

but this full update can oscillate. A smaller step size, or a backtracking line search on the outer objective, gives a more robust method.

A suitable stationarity residual is

$$\max_i \left\| u_i^{(k)} - \left(x_i - \bar{y}_i^{(k)} \right) \right\|.$$

It should be monitored together with the marginal errors

$$\left\| P^{(k)} \mathbf{1}_m - \mathbf{a} \right\|, \quad \left\| (P^{(k)})^\top \mathbf{1}_n - \mathbf{b} \right\|.$$

For small ε , the entries of K can overflow or underflow. As in standard entropic optimal transport, one should then implement the Sinkhorn iterations in the logarithmic domain using log-sum-exp stabilization.

8 Summary

The main conclusions are:

- The barycentric weak transport objective is convex in the coupling P , because it is a sum of convex quadratic functions of row barycenters.
- It is generally not strictly convex, since the objective cannot distinguish couplings having the same conditional barycenters.
- Adding $\varepsilon \text{KL}(P \| \mathbf{a} \mathbf{b}^\top)$ makes the discrete objective strictly convex and selects a unique coupling.
- The regularized problem is not a standard entropic OT problem with a fixed cost matrix: its effective cost depends on P .
- A Fenchel saddle-point representation introduces variables u_i for which the coupling subproblem becomes an ordinary entropic transport problem. It can therefore be solved by an inner Sinkhorn scaling, combined with an outer damped ascent or fixed-point iteration.

9 Linearization and Sinkhorn iteration

Let

$$b_\pi(x) := \int_{\mathbb{R}^d} y \pi_x(dy)$$

be the barycentric projection of a coupling $\pi(dx, dy) = \alpha(dx) \pi_x(dy)$. Define the coupling-dependent cost

$$c_\pi(x, y) := 2 \langle x - b_\pi(x), x - y \rangle.$$

Then the quadratic weak transport functional

$$\mathcal{B}(\pi) = \int_{\mathbb{R}^d} \|x - b_\pi(x)\|^2 \alpha(dx)$$

satisfies the exact identity

$$\begin{aligned} \frac{1}{2} \langle c_\pi, \pi \rangle &= \frac{1}{2} \int_{\mathbb{R}^d \times \mathbb{R}^d} c_\pi(x, y) \pi(dx, dy) \\ &= \int_{\mathbb{R}^d} \left\langle x - b_\pi(x), x - \int y \pi_x(dy) \right\rangle \alpha(dx) \\ &= \int_{\mathbb{R}^d} \|x - b_\pi(x)\|^2 \alpha(dx). \end{aligned}$$

Therefore,

$$\mathcal{B}(\pi) = \text{cst} + \frac{1}{2} \langle c_\pi, \pi \rangle, \quad \text{cst} = 0.$$

The actual first-variation cost is

$$\ell_\pi(x, y) = 2 \langle b_\pi(x) - x, y \rangle.$$

The two costs differ only by a function of x :

$$c_\pi(x, y) = \ell_\pi(x, y) + 2 \langle x - b_\pi(x), x \rangle.$$

Since every admissible coupling has first marginal α , adding a function of x to an OT cost does not change the minimizing coupling. Thus, c_π can be used as the linearized OT cost.

More precisely, for any $\gamma \in \Pi(\alpha, \beta)$,

$$\mathcal{B}(\gamma) = \mathcal{B}(\pi) + \langle c_\pi, \gamma - \pi \rangle + \int \|b_\gamma(x) - b_\pi(x)\|^2 \alpha(dx).$$

Hence the first-order approximation around π is

$$\mathcal{B}(\gamma) \simeq \mathcal{B}(\pi) + \langle c_\pi, \gamma - \pi \rangle.$$

9.1 Discrete expression

For

$$\alpha = \sum_{i=1}^n a_i \delta_{x_i}, \quad \beta = \sum_{j=1}^m b_j \delta_{y_j},$$

let $P \in U(\mathbf{a}, \mathbf{b})$ and define

$$\bar{y}_i(P) = \frac{1}{a_i} \sum_{j=1}^m P_{ij} y_j.$$

The discrete coupling-dependent cost matrix is

$$\boxed{C_{ij}(P) = 2 \langle x_i - \bar{y}_i(P), x_i - y_j \rangle.}$$

It satisfies

$$\begin{aligned} \frac{1}{2} \langle C(P), P \rangle &= \frac{1}{2} \sum_{i,j} P_{ij} C_{ij}(P) \\ &= \sum_i a_i \langle x_i - \bar{y}_i(P), x_i - \bar{y}_i(P) \rangle \\ &= \sum_i a_i \|x_i - \bar{y}_i(P)\|^2. \end{aligned}$$

Consequently,

$$\boxed{\mathcal{B}(P) = \frac{1}{2} \langle C(P), P \rangle.}$$

The entries can equivalently be written as

$$C_{ij}(P) = 2 \left\langle \frac{1}{a_i} \sum_{\ell=1}^m P_{i\ell} (y_\ell - x_i), y_j - x_i \right\rangle.$$

9.2 Linearized entropic OT subproblem

Given the current coupling $P^{(k)}$, compute

$$C_{ij}^{(k)} := C_{ij}(P^{(k)}).$$

The linearized KL-regularized problem is the classical entropic OT problem

$$Q^{(k)} = \operatorname{argmin}_{Q \in U(\mathbf{a}, \mathbf{b})} \left\{ \langle C^{(k)}, Q \rangle + \varepsilon \operatorname{KL} \left(Q \parallel \mathbf{a} \mathbf{b}^\top \right) \right\}.$$

Introduce the Gibbs kernel

$$K_{ij}^{(k)} = \exp \left(-\frac{C_{ij}^{(k)}}{\varepsilon} \right).$$

The solution has the scaling form

$$Q_{ij}^{(k)} = a_i b_j u_i^{(k)} K_{ij}^{(k)} v_j^{(k)}.$$

The scaling vectors are computed with standard Sinkhorn iterations:

$$u_i \leftarrow \frac{1}{\sum_{j=1}^m b_j K_{ij}^{(k)} v_j}, \quad v_j \leftarrow \frac{1}{\sum_{i=1}^n a_i u_i K_{ij}^{(k)}}.$$

9.3 Linearize-and-Sinkhorn algorithm

A simple fixed-point iteration consists of setting

$$P^{(k+1)} = Q^{(k)}.$$

A more stable version uses a relaxed update

$$P^{(k+1)} = (1 - \tau_k) P^{(k)} + \tau_k Q^{(k)}, \quad 0 < \tau_k \leq 1.$$

The complete iteration is therefore:

1. $\bar{y}_i^{(k)} = \frac{1}{a_i} \sum_j P_{ij}^{(k)} y_j,$
2. $C_{ij}^{(k)} = 2 \left\langle x_i - \bar{y}_i^{(k)}, x_i - y_j \right\rangle,$
3. $Q^{(k)} = \operatorname{Sinkhorn} \left(\mathbf{a}, \mathbf{b}, \exp(-C^{(k)}/\varepsilon) \right),$
4. $P^{(k+1)} = (1 - \tau_k) P^{(k)} + \tau_k Q^{(k)}.$

This is an entropic generalized conditional-gradient algorithm. The quadratic barycentric functional is linearized at the current coupling, and the resulting linear entropic OT problem is solved by Sinkhorn.

The full step $\tau_k = 1$ gives the simplest fixed-point algorithm, but it may oscillate. Damping, for example

$$\tau_k = \frac{2}{k+2},$$

or an objective-based line search generally gives a more robust iteration.